

정삼광 → 조갑

① 공간에 있는 물체를 잡는다.

2021.11.23 설재훈 2025.11.04

클래스 데이터 → Unity → 물체 잡고 → 종류 → 물리적 판박

function

1. cross - feedback

① tendon system

: 유압 아 모터 등으로 당겨서 손가락에 인장력을 전달하여 제어력 인가

- 장점

- 경량성: 구동 장치가 가벼움
- 설치 및 개조가 용이

- 단점

- 인장력이 약하고, 압축력을 주기 힘들
- 케이블 장력의 비선형, 마찰로 인해 제어 정밀도 ↓

↳ tendon이 지나가는 line을 정확하게 설계해야됨

안그러면 tendon을 당기면 최소 경로로 이탈하여 정밀도 ↓

그림

5분

② frame system

: 손 자체에 액추에이터나 강성 프레임(frame)을 사용하여 밀고 당기는 방식으로 제어력 인가

- 장점

- 인장력, 압축력을 강직하게 전달할 수 있음
- 상대적 높은 정밀도

- 단점

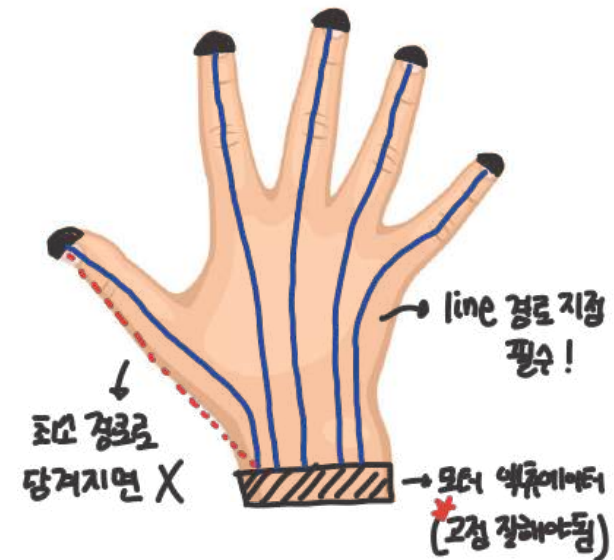
- 액추에이터, 프레임 자체가 크고 무거움
- 비용 ↑, 시간 ↑

↳ 우리 목적과 맞지 않음

조갑이 필요한 이유

조갑으로 구현한지

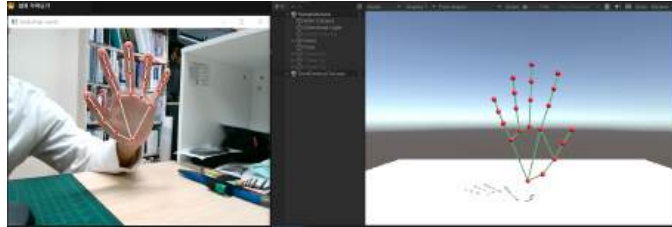
도면, 프로그래밍 → sketch



2. motion tracking

① vision 기반 tracking

: 카메라와 컴퓨터 머신러닝을 사용하여 위치, 자세 정보 추적



: 손가락 21개 관절의 3D 좌표 → 시뮬레이션 엔진

② 센서 기반 tracking

: 손가락 관절에 부착된 센서를 통해 물리장을 측정하여 움직임 예측

- 저항성 필라멘트

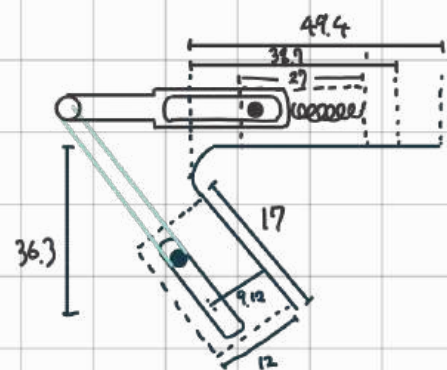
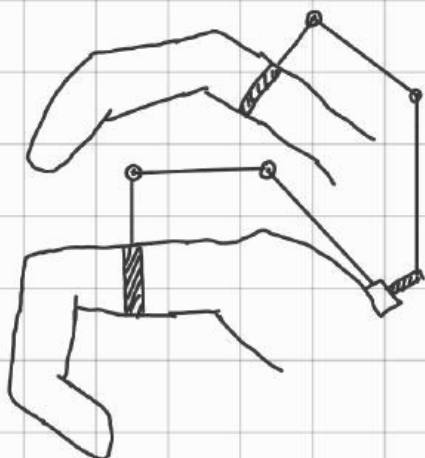
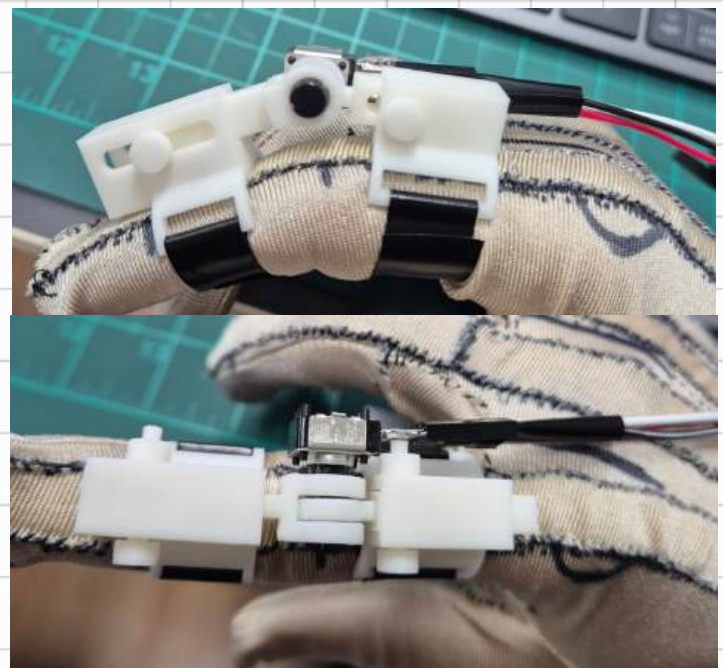
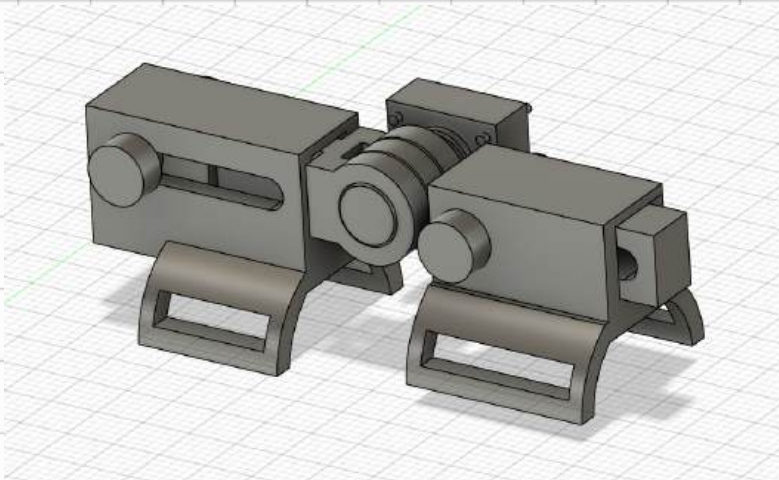
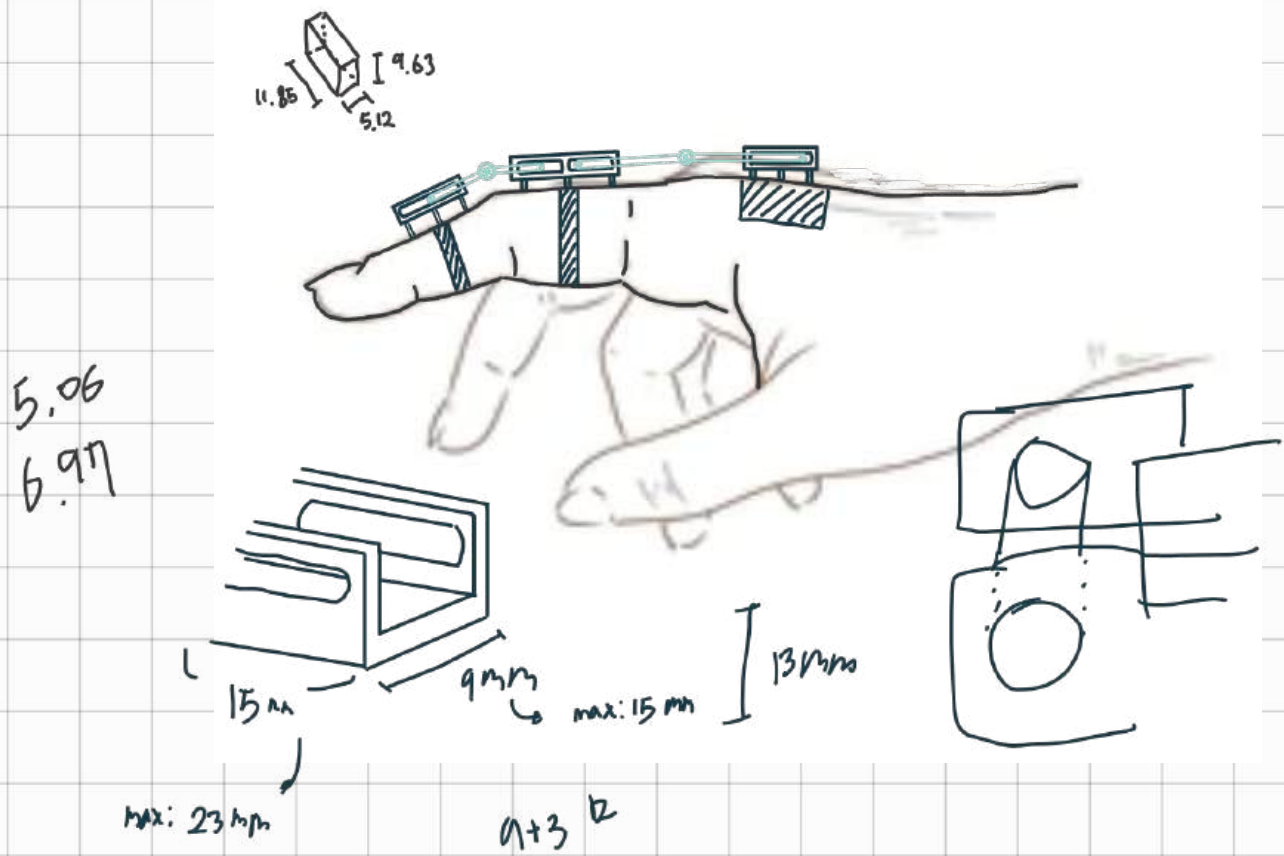
→ ————— ← 압축 ⇒ 저항 ↓

→ ————— ← 인장 ⇒ 저항 ↑

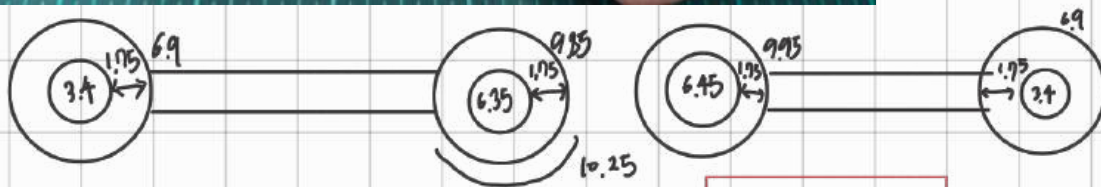
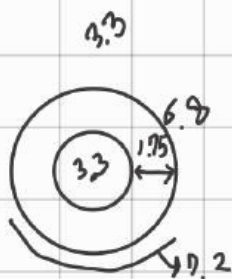
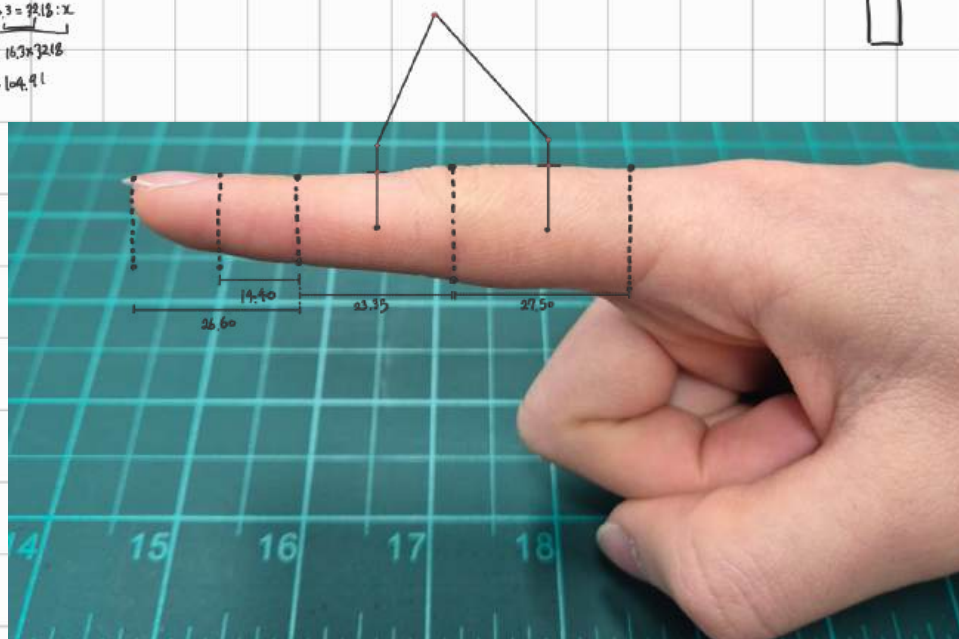
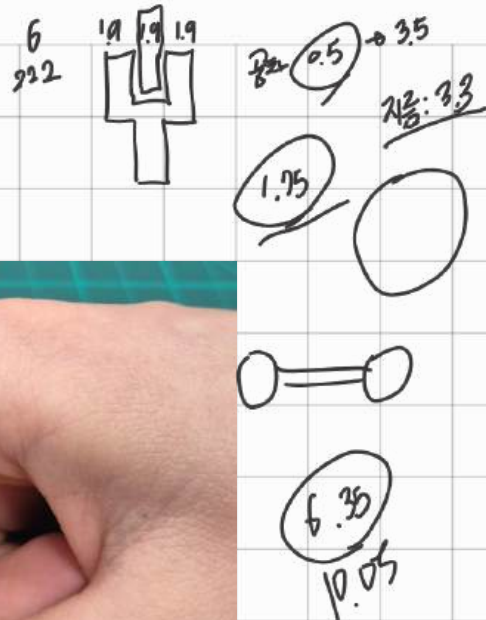
3. manipulator

: 손 관절의 3D 좌표 → 로봇 손 전달 후 로봇 팔 각 관절의 좌절 각도 계산

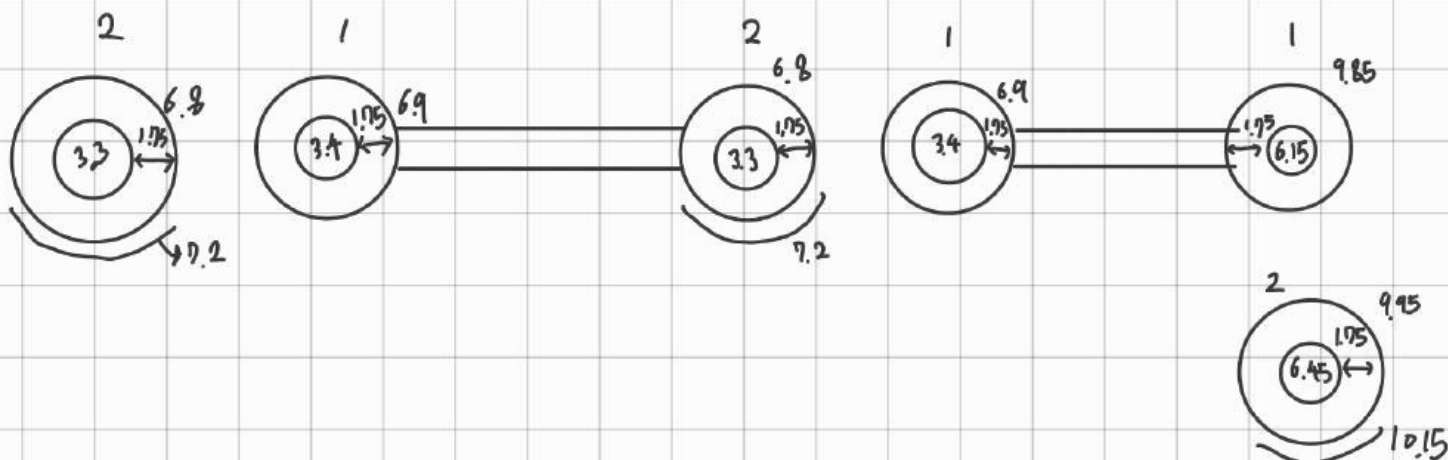
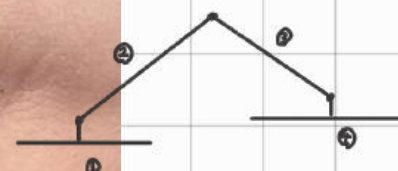
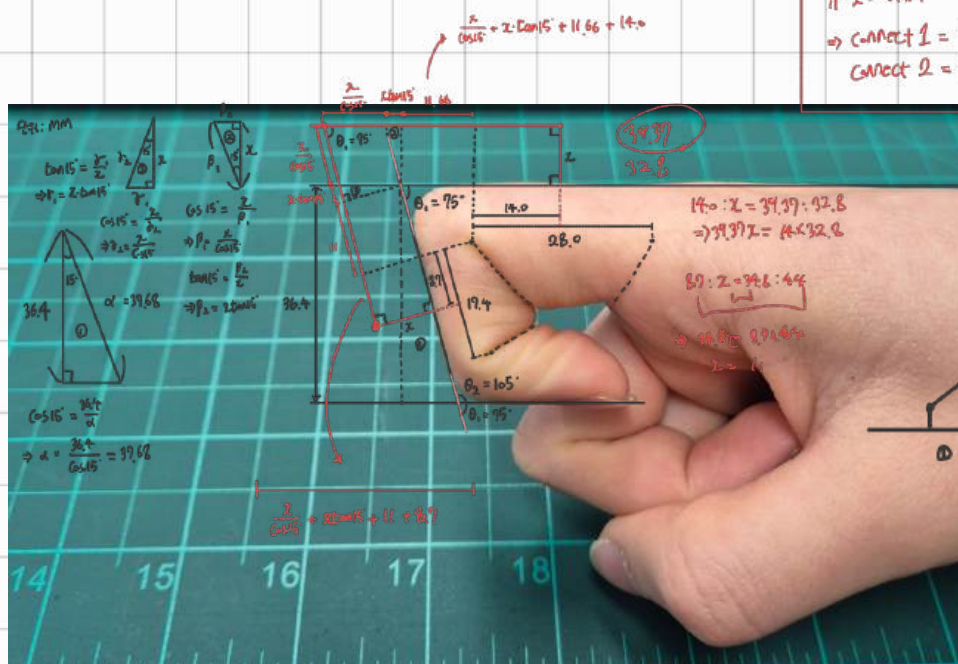
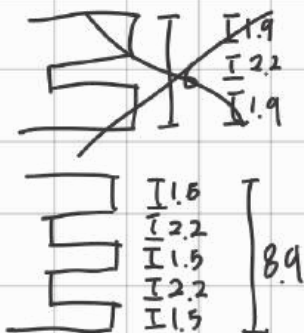
- ↳ . 관절 한계
 . 장애물 회피
 . 특이점 회피

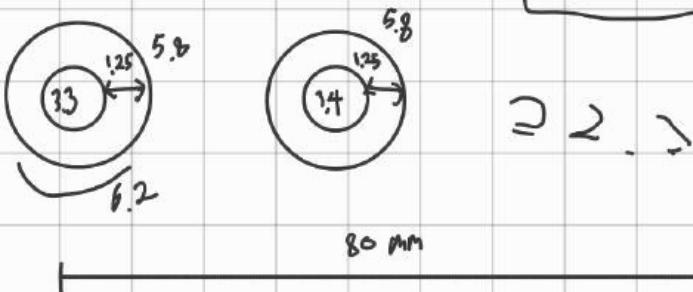
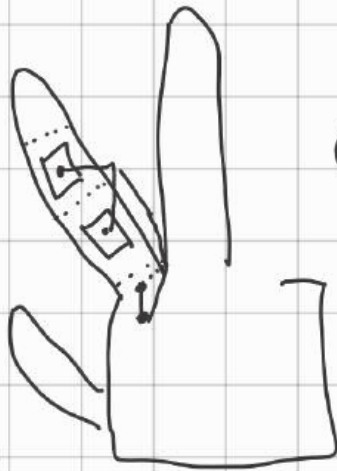
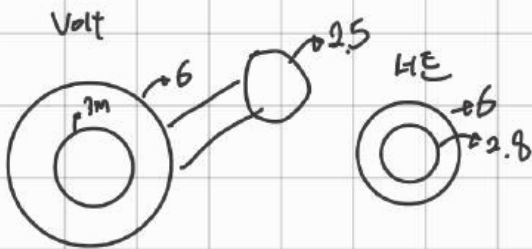
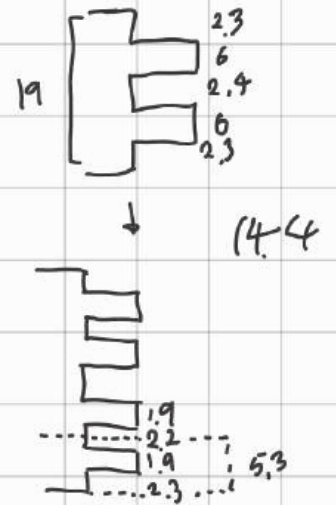
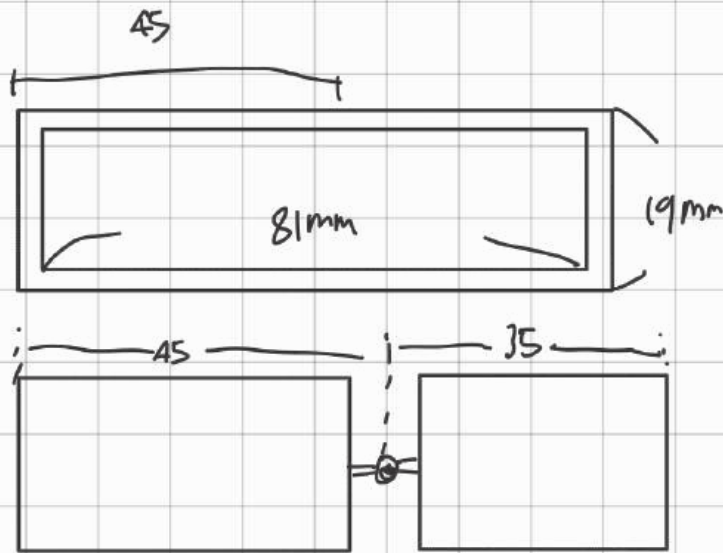


$5.45 \rightarrow 6.35$ 20% 20%
 $3 \rightarrow 3.3$ 10% 6.5 3.4

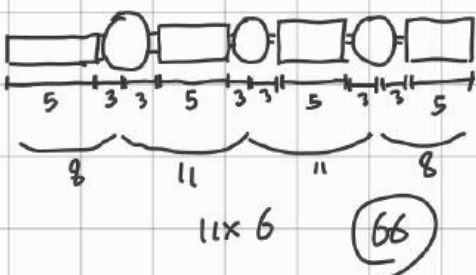


if $x = 5 \text{ mm}$
 $\Rightarrow \text{correct 1} = 26.22$
 $\text{correct 2} = 32.18$



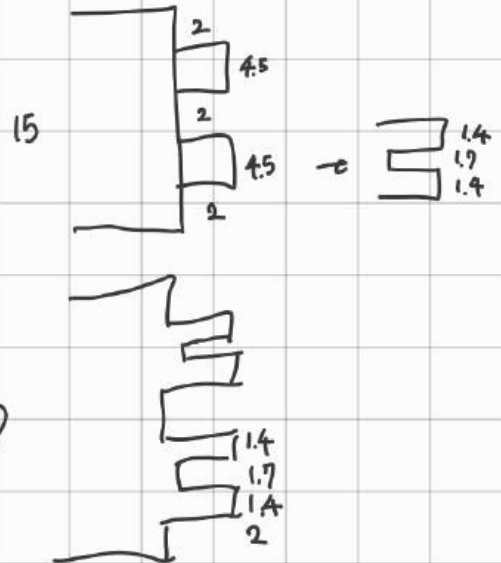


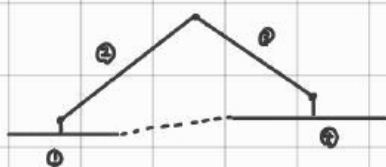
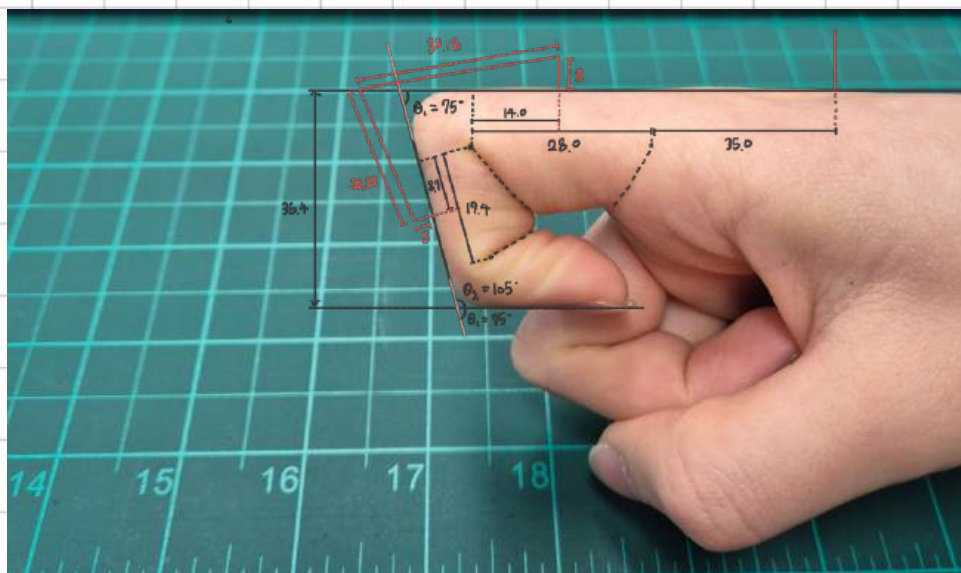
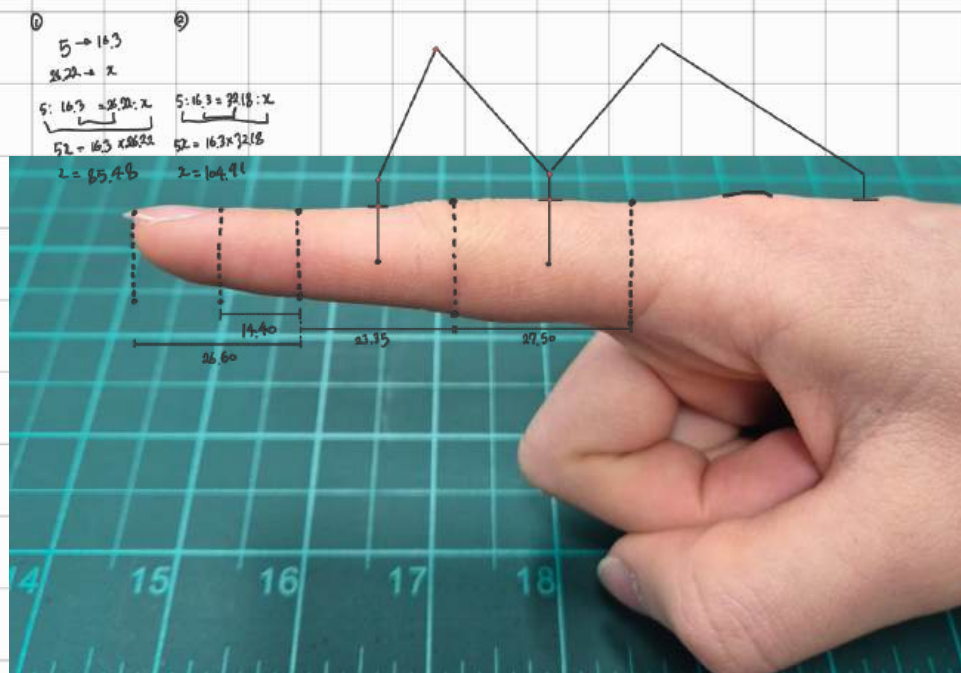
22.2



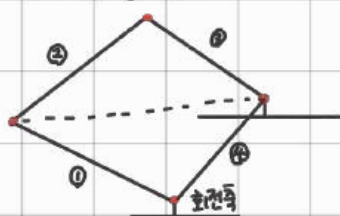
#2 = 81 (17)

14.4

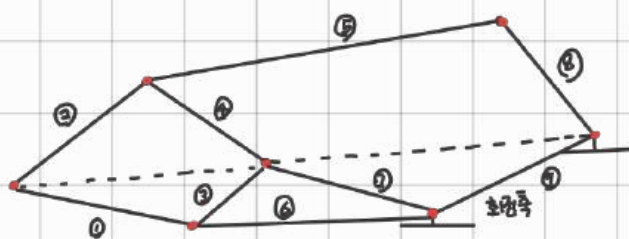




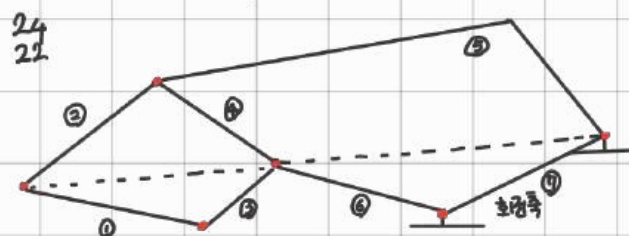
$$M = 3(3-1) - 2 \times 3 = 0 \text{ dof}$$



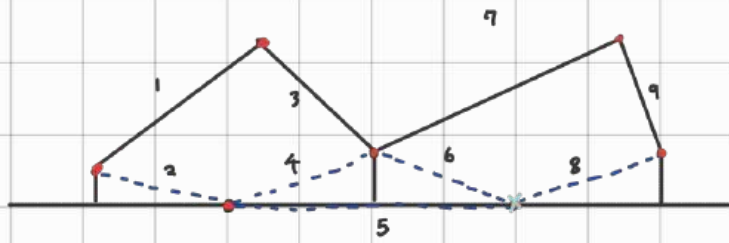
$$M = 3(4-1) - 2 \times 4 = 1 \text{ dof}$$



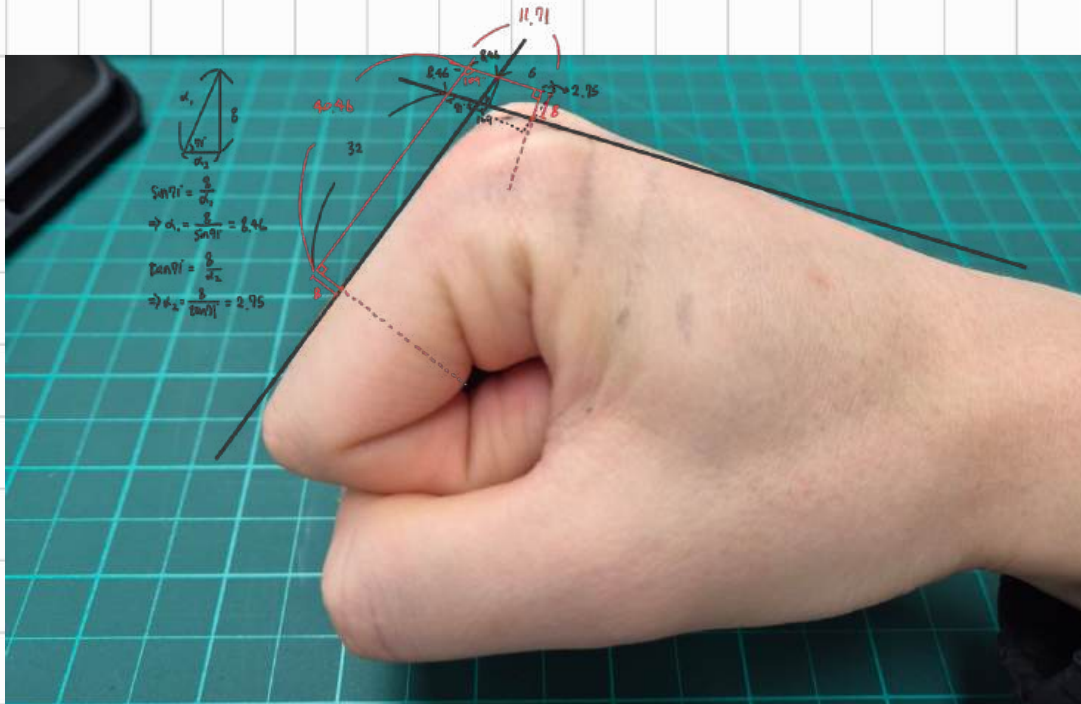
$$M = 3 \times (9-1) - 2 \times 11 = 2 \text{ dof}$$



$$M = 3 \times (7-1) - 2 \times 8 = 2 \text{ dof}$$



$$M = 3 \times (9 - 1) - 2 \times 11 = 2 \text{ dof}$$

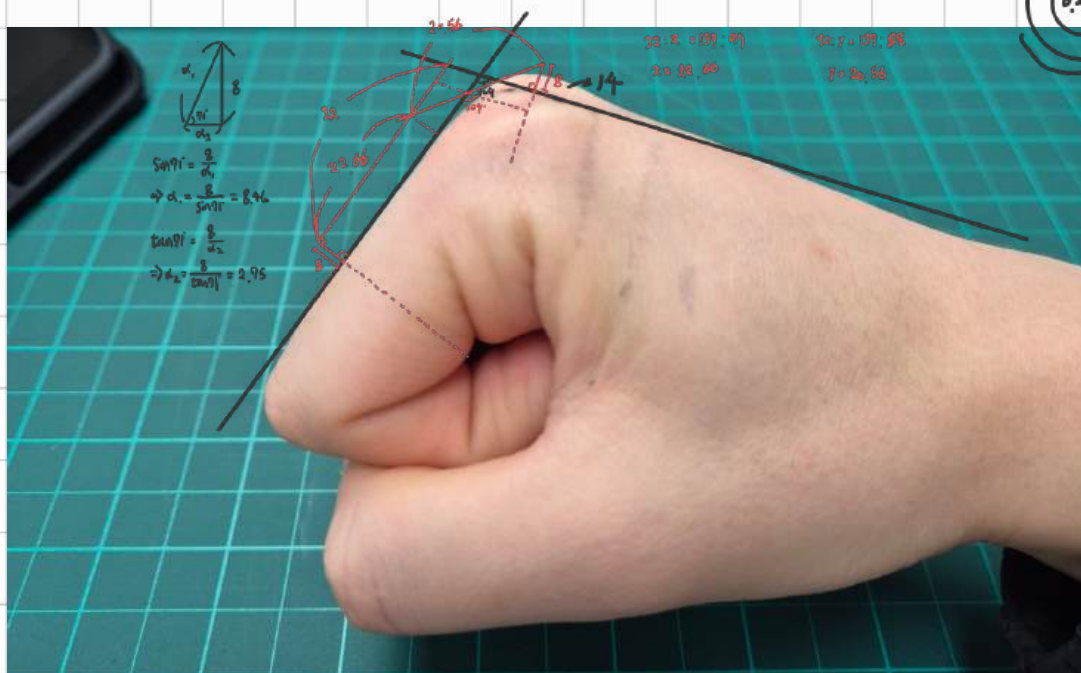
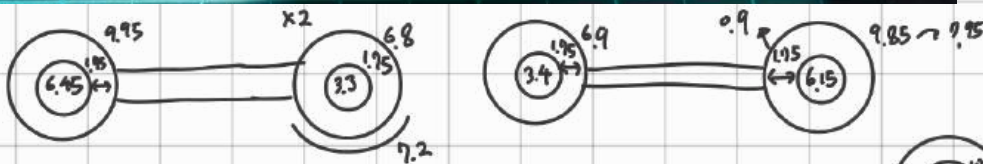


$$\sin \theta_1 = \frac{8}{a_1}$$

$$\Rightarrow a_1 = \frac{8}{\sin \theta_1} = 8.46$$

$$\tan \theta_1 = \frac{8}{a_2}$$

$$\Rightarrow a_2 = \frac{8}{\tan \theta_1} = 2.95$$



$$\sin \theta_1 = \frac{8}{a_1}$$

$$\Rightarrow a_1 = \frac{8}{\sin \theta_1} = 8.46$$

$$\tan \theta_1 = \frac{8}{a_2}$$

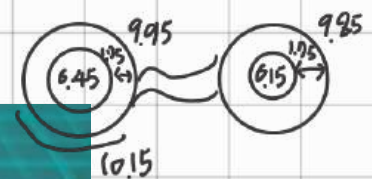
$$\Rightarrow a_2 = \frac{8}{\tan \theta_1} = 2.95$$

$$30.4 \pm 0.07, 0.07$$

$$20.23, 0.07$$

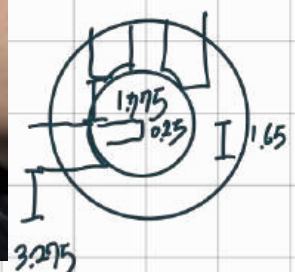
$$30.7 \pm 0.07, 0.07$$

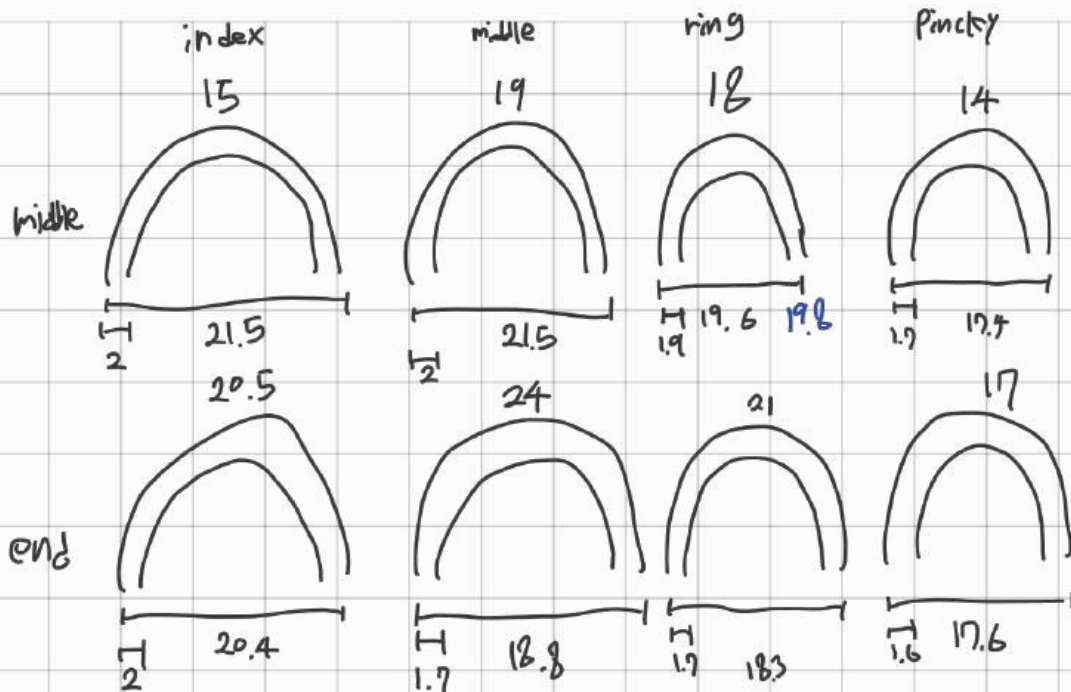
$$20.23, 0.07$$



6.15

6.017





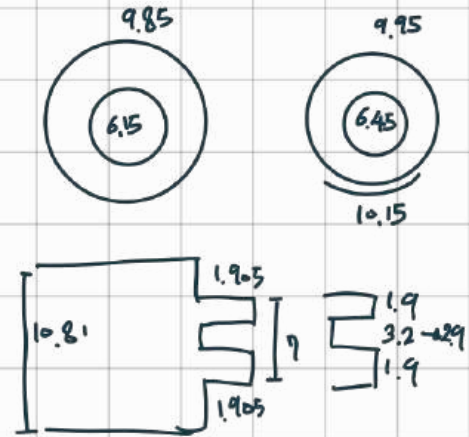
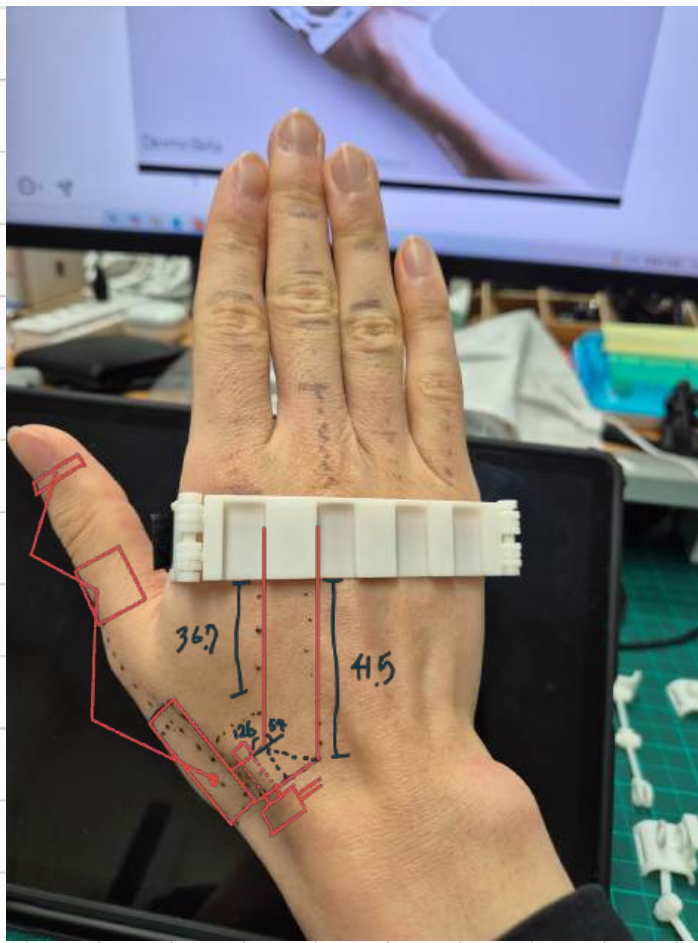
inter		middle		ring		pinky	
m-c1	동결	m-c1	동결	m-c1	-2	m-c1	동결
m-c2	동결	m-c2	-2	m-c2	-2	m-c2	동결
e-c1	동결	e-c1	동결	e-c1	+2	e-c1	동결
e-c2	동결	e-c2	동결	e-c2	+2	e-c2	동결

thumb

m-c1 동결
m-c2 동결
e-c1 +2
e-c2 동결

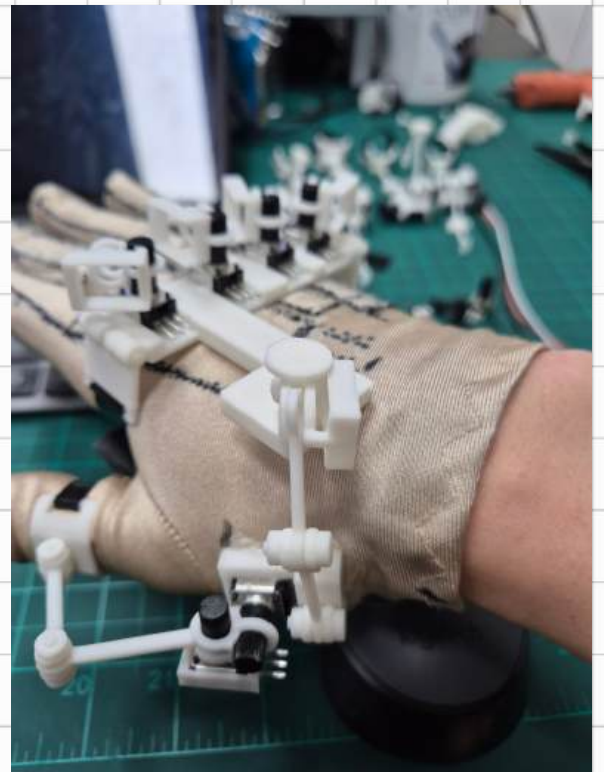
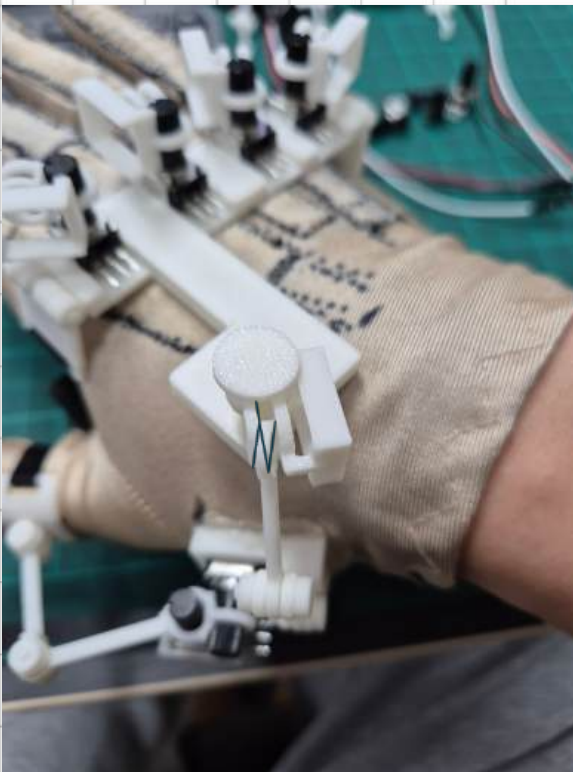
ball 길이 포함

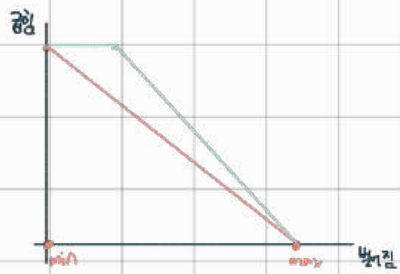
15.03
12.03 + 3
15.03



마그네틱 브래킷

모터알고 저항 부품

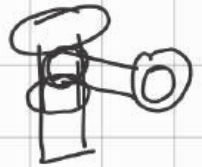




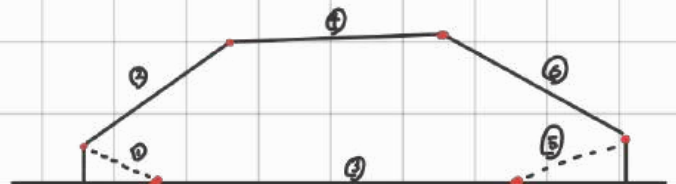
구조

핀

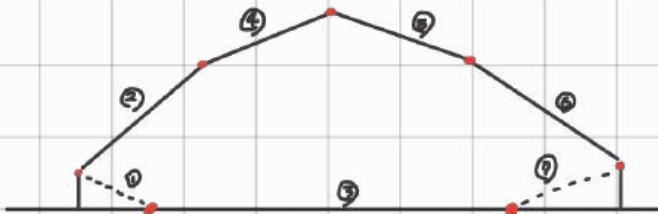
반점



$$M = 3(5-1) - 2 \times 5 = 2 \text{ dof}$$



$$M = 3(6-1) - 2 \times 6 = 3 \text{ dof}$$



$$M = 3(7-1) - 2 \times 7 = 4 \text{ dof}$$

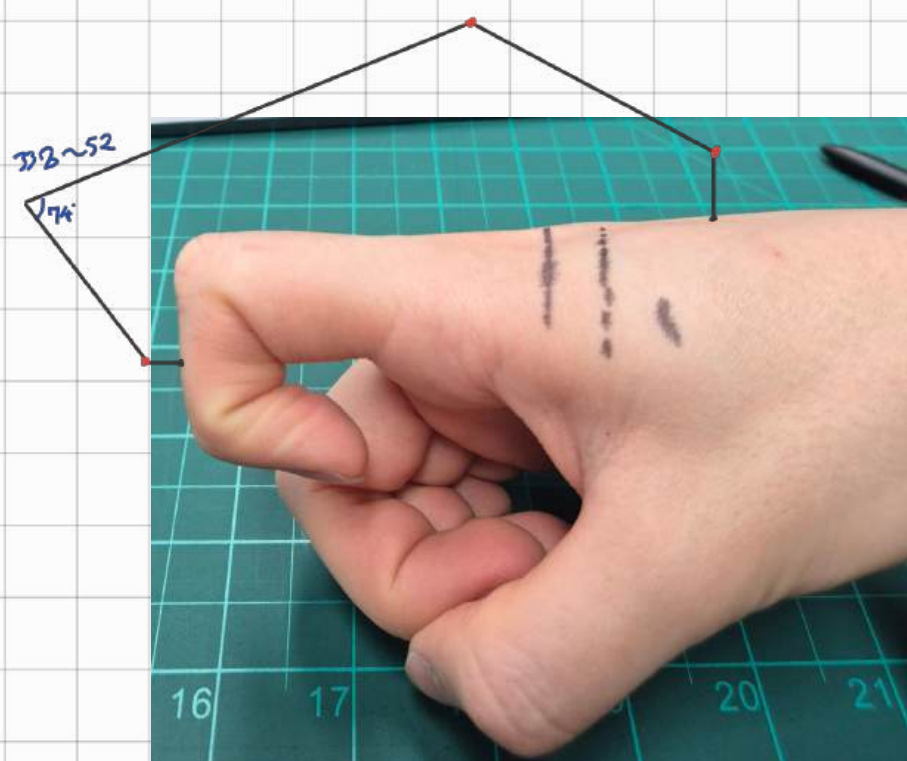
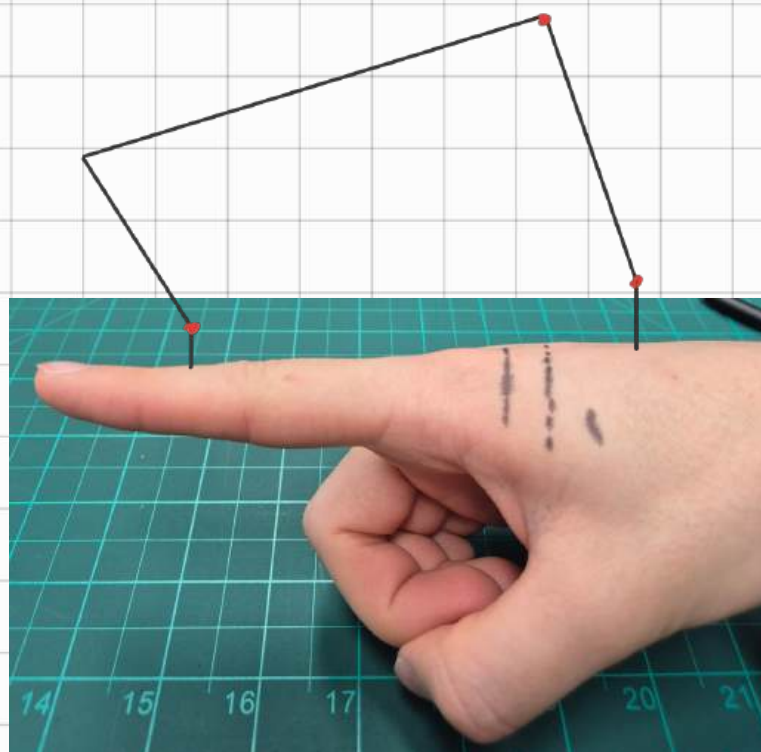
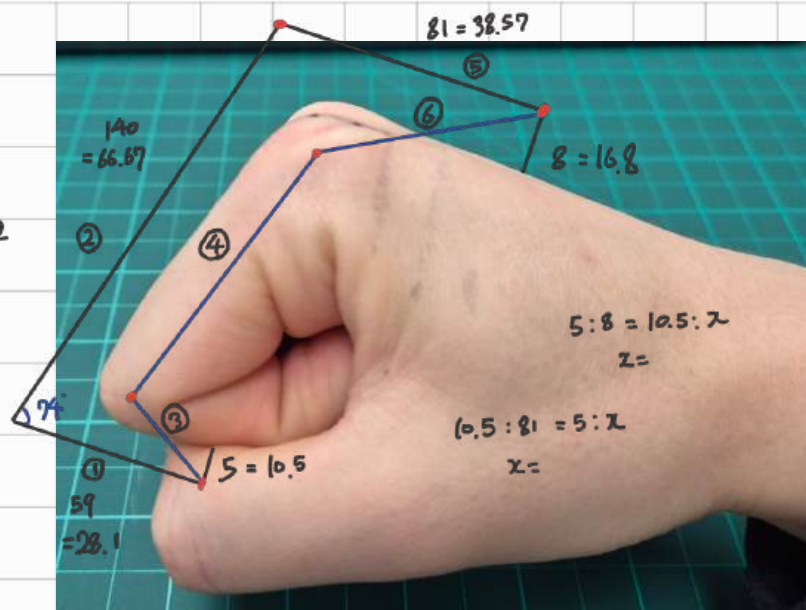


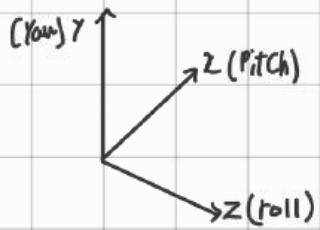
$$M = 3(4-1) - 2 \times 4 = 1 \text{ dof}$$



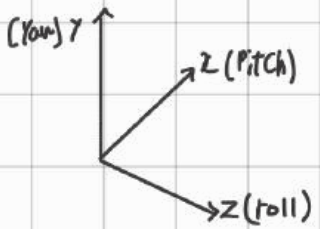
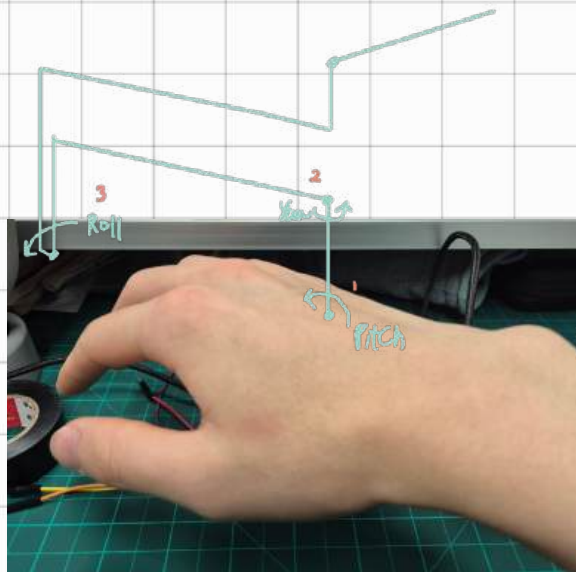
$$M = 3(6-1) - 2 \times 6 = 3$$

$$n = 3(5-1) - 2 \times 5 = 2$$

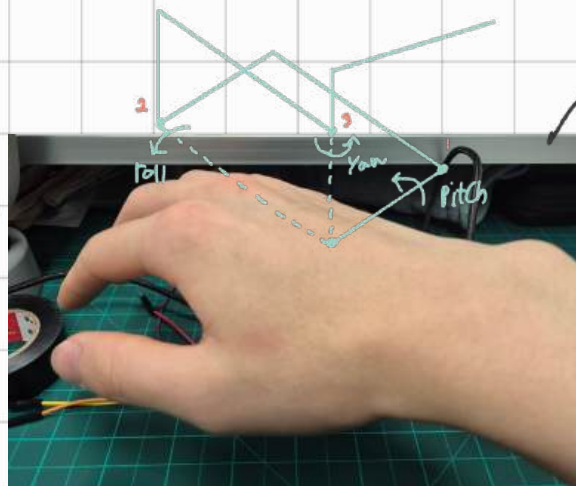




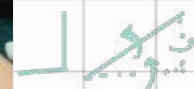
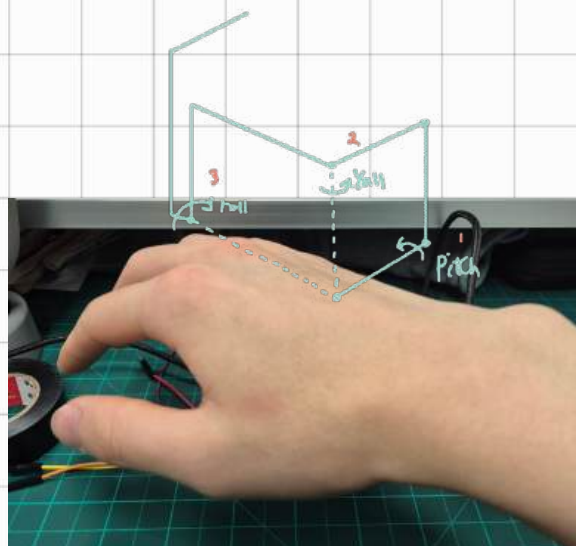
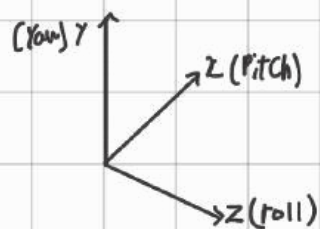
프레임 → Roll → Yaw → Pitch → 끝

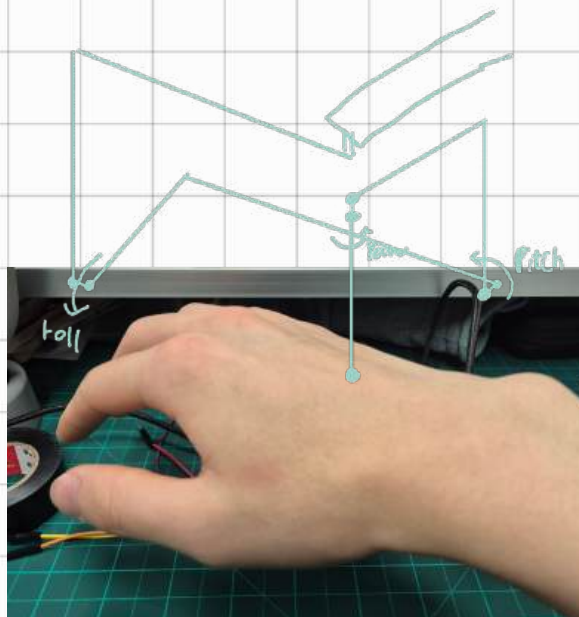
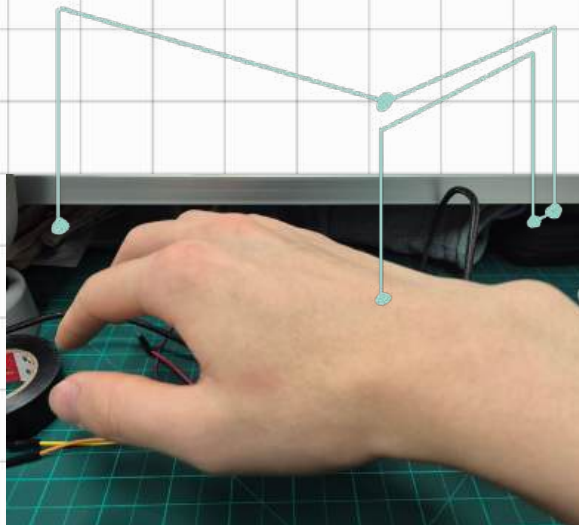
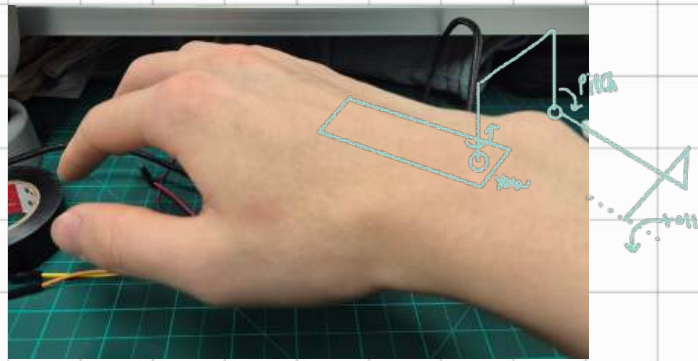


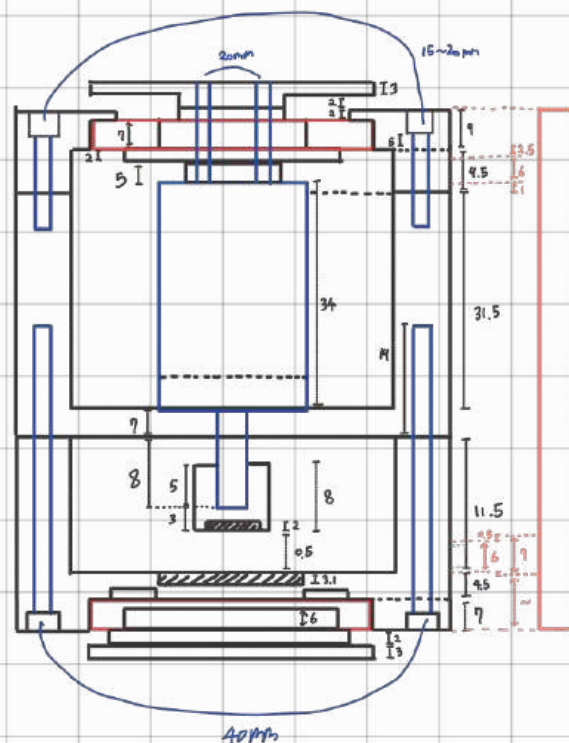
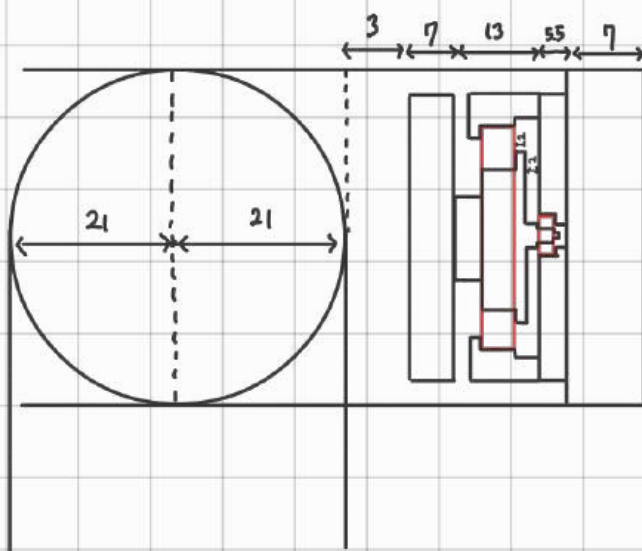
프레임 → Yaw → Roll → Pitch → 끝



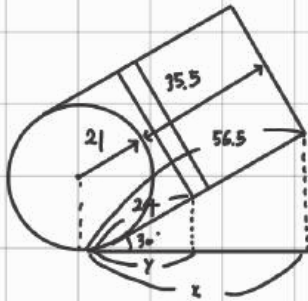
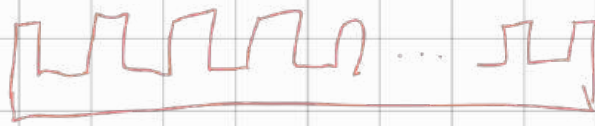
→ Roll = 90° 일때 김벌라







26.581 42.531

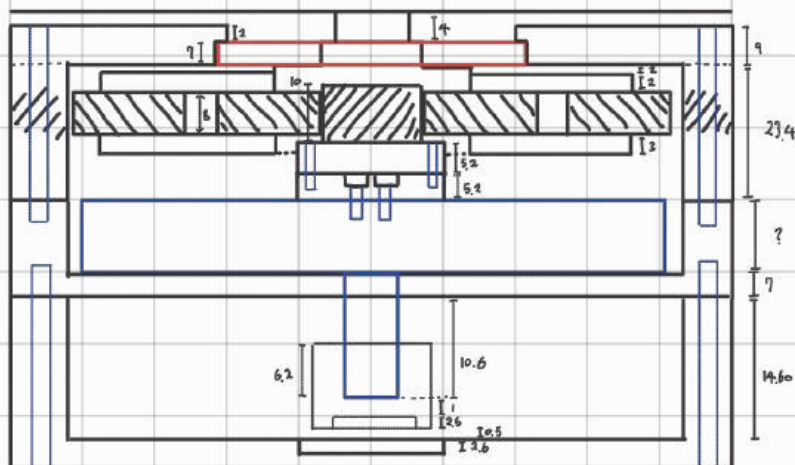


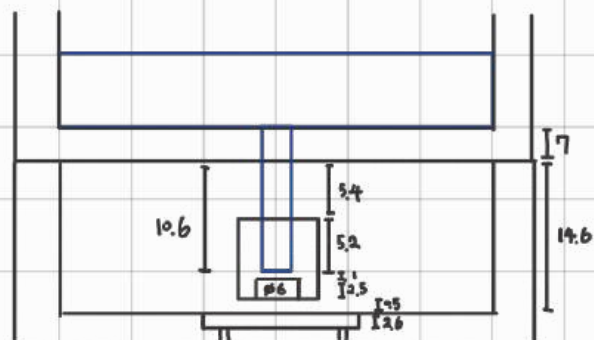
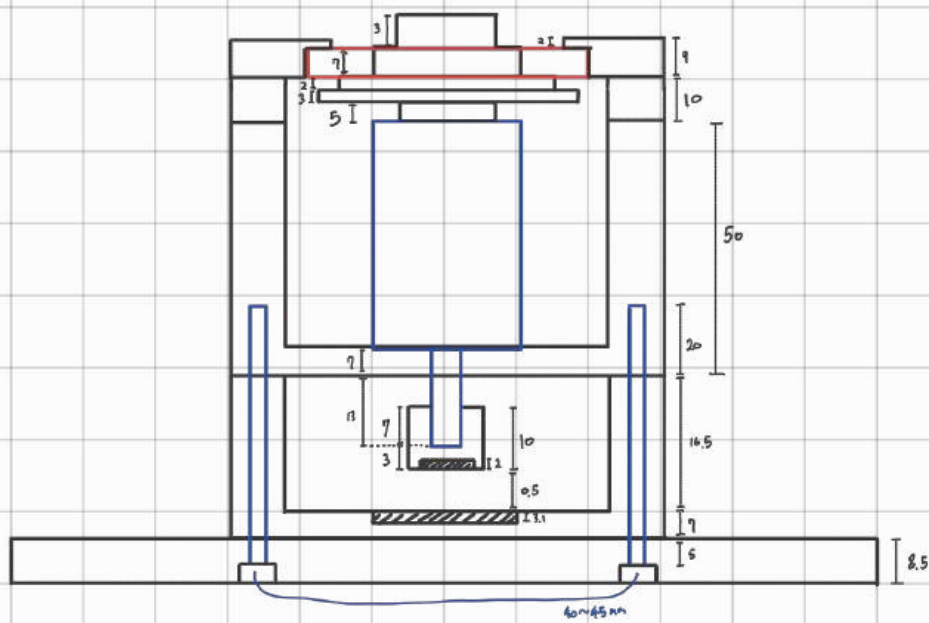
$$\cos 30^\circ = \frac{x}{56.5}$$

$$x = 56.5 \times \cos 30^\circ = 48.93$$

$$y = 21 \times \cos 30^\circ = 20.79$$

14:1 이면 폭 14m?

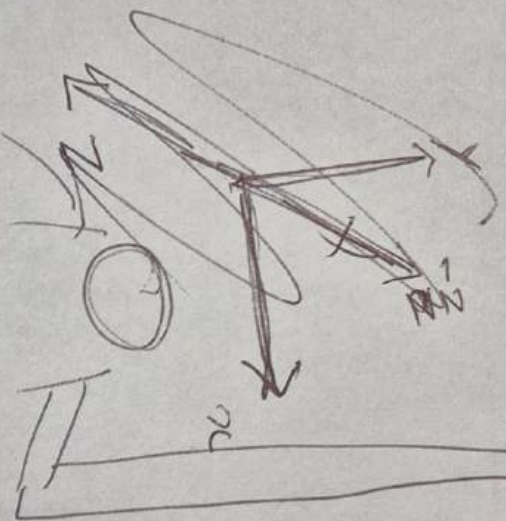
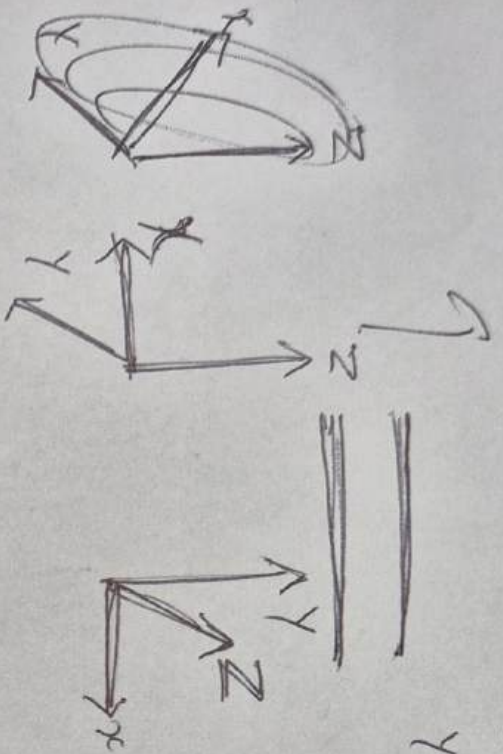




$$x = -x$$

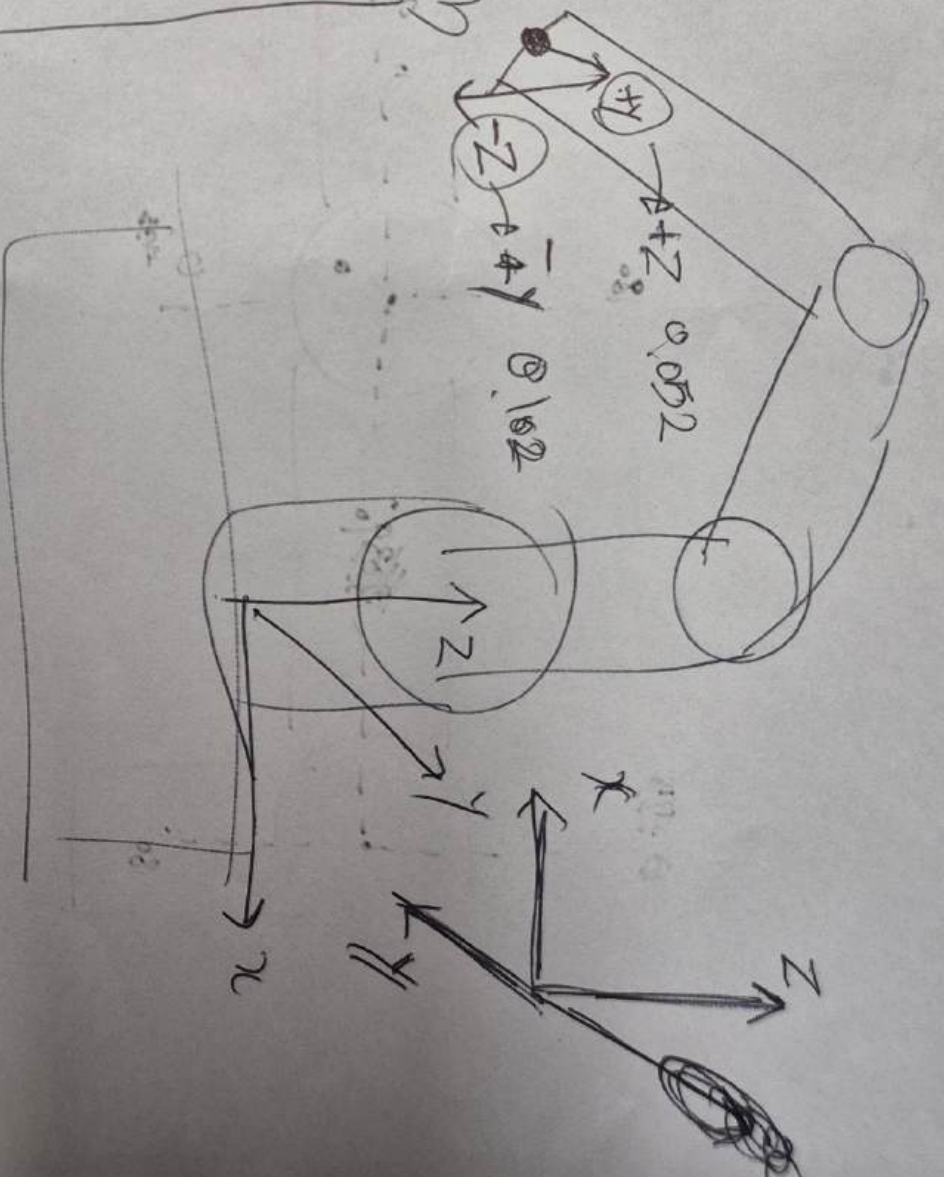
$$z = y$$

$$y = z$$



$$z \rightarrow y$$

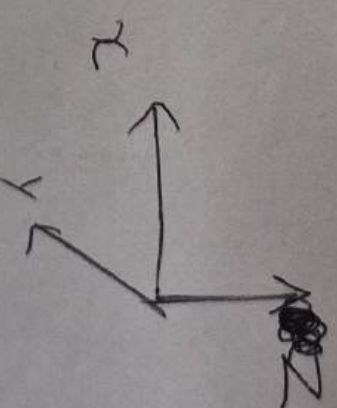
$$y \rightarrow -z$$



$$x \rightarrow -x$$

$$y \rightarrow -y$$

$$z \rightarrow -z$$

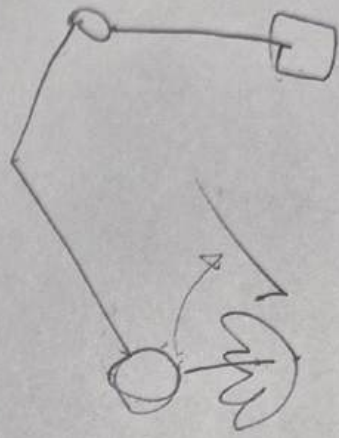


1/2 주를 때

band world

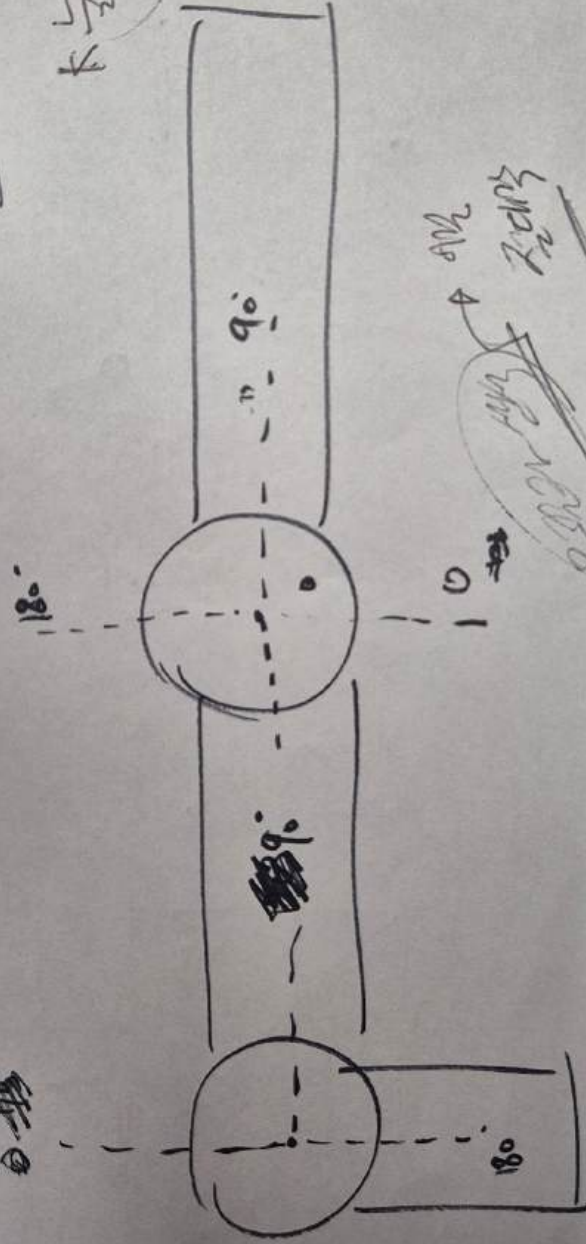
=

band world



band world

band world



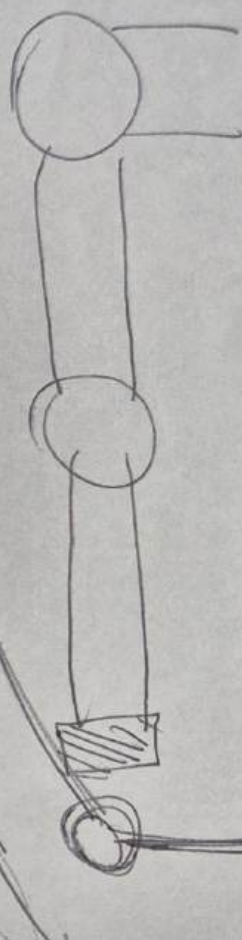
band world

band world

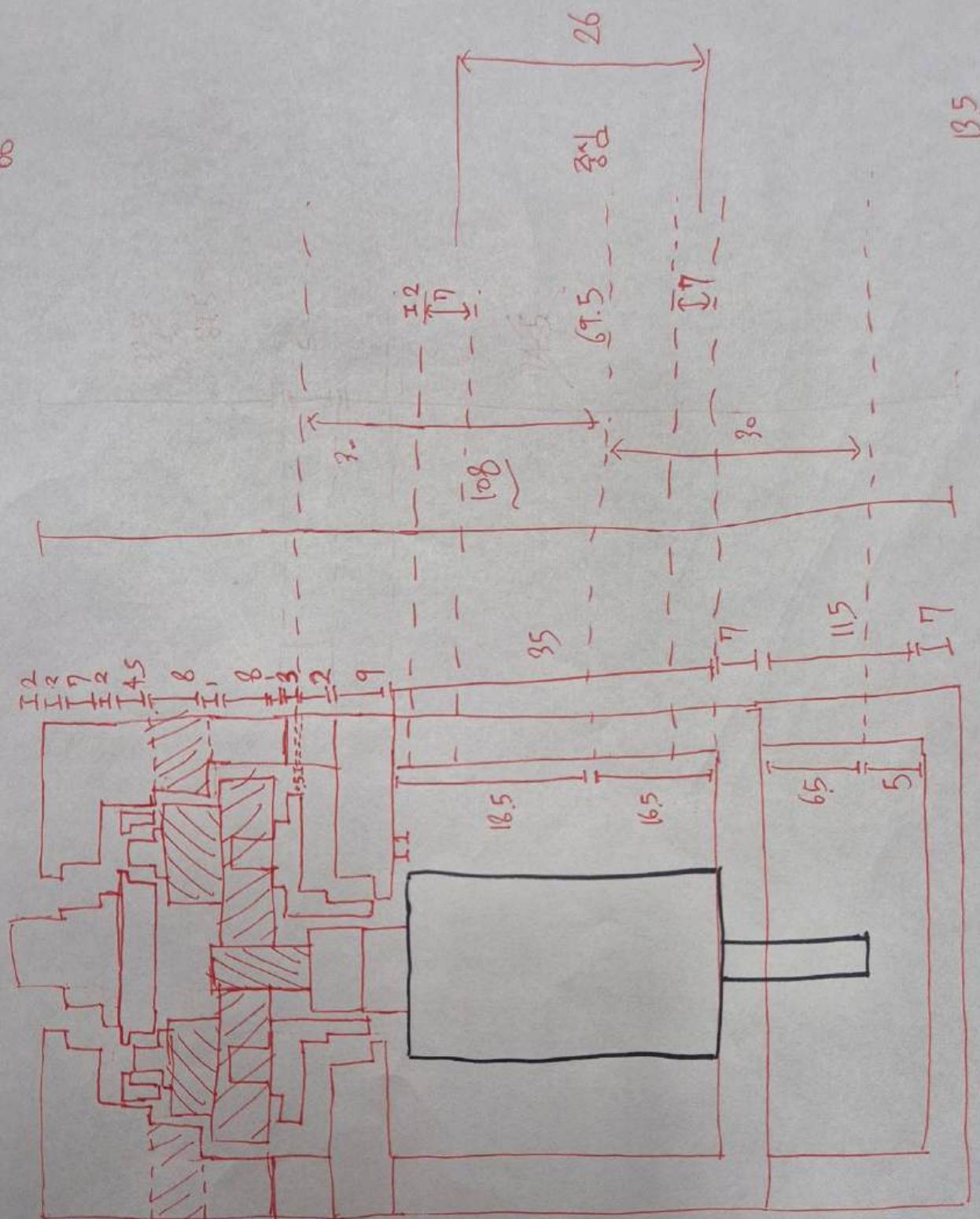
band world

band world

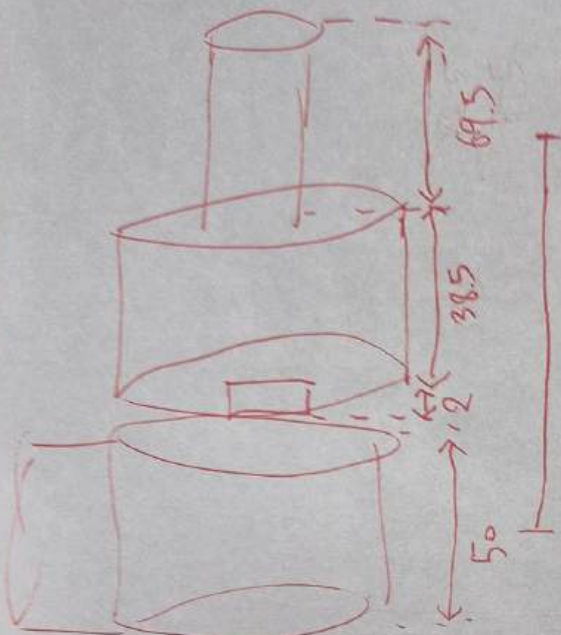
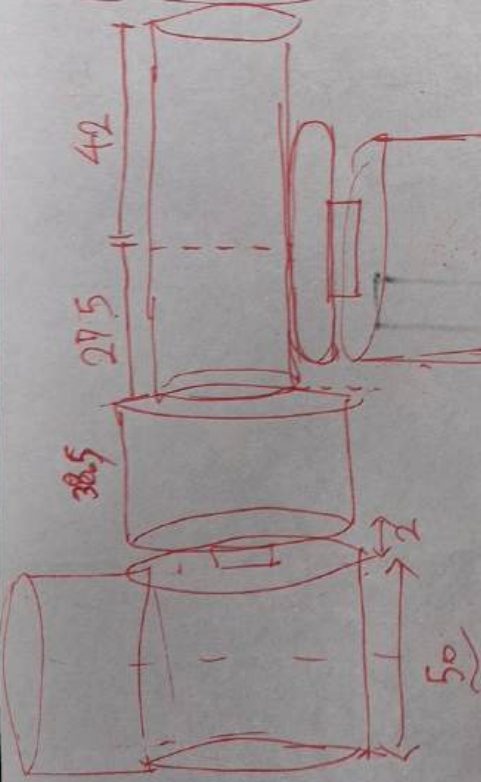
band world



band world



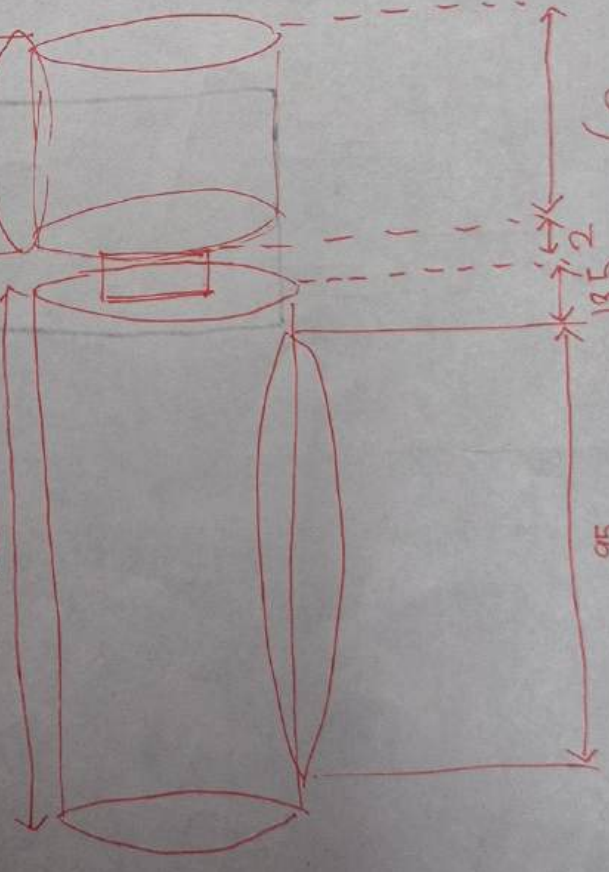
5.4 11+2



$$25 + 2 + 38.5 + x = 93$$

$$\therefore x = 27.5$$

102



66

$$25 + 2 + x = 93$$

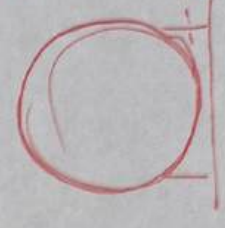
900

9072

$$45 + x + 3 + 45$$

4

45



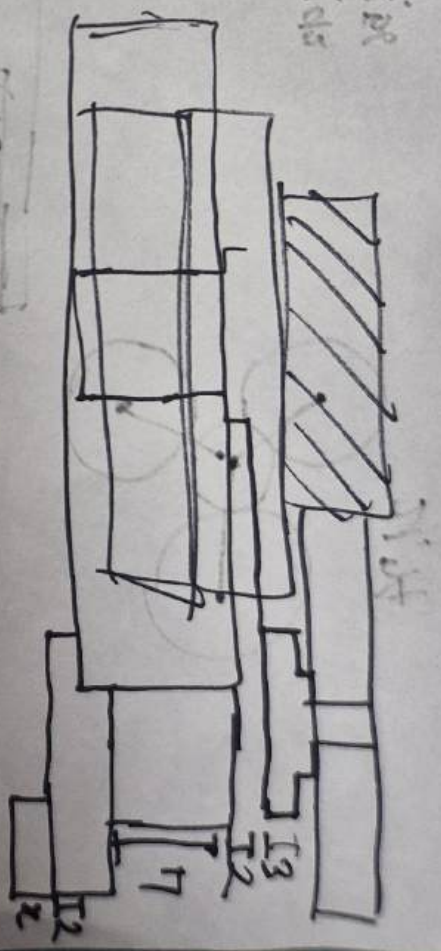
13

$$45 + 2 + 3 + 45$$

93

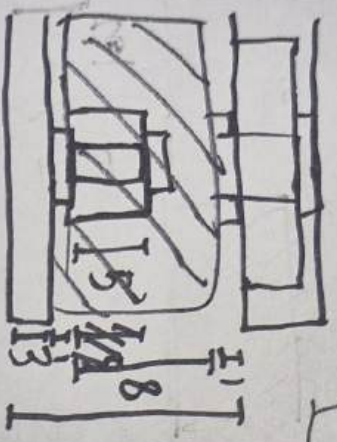


ଗୋଟିଏ ଗଡ଼ : ୧୦' ଗୋଟିଏ ଗଡ଼ : ୧୦' ଗୋଟିଏ ଗଡ଼ : ୧୦' ଗୋଟିଏ ଗଡ଼ : ୧୦'
 ଗୋଟିଏ ଗଡ଼ : ୧୦' ଗୋଟିଏ ଗଡ଼ : ୧୦' ଗୋଟିଏ ଗଡ଼ : ୧୦' ଗୋଟିଏ ଗଡ଼ : ୧୦'
 ଗୋଟିଏ ଗଡ଼ : ୧୦' ଗୋଟିଏ ଗଡ଼ : ୧୦' ଗୋଟିଏ ଗଡ଼ : ୧୦' ଗୋଟିଏ ଗଡ଼ : ୧୦'



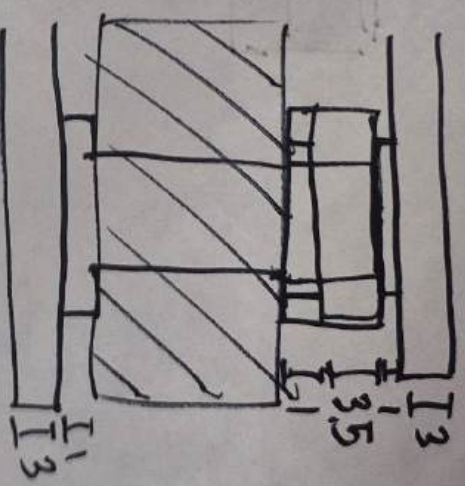
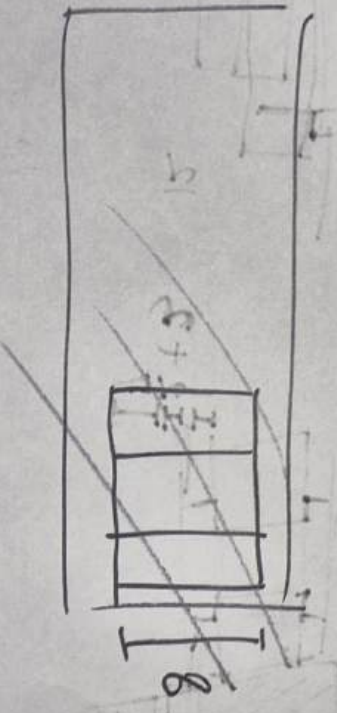
$$3+2+7+2 = 14$$

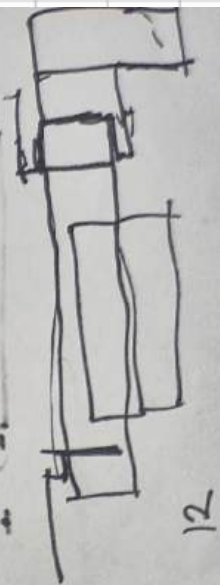
$$14-2-17-2-3$$



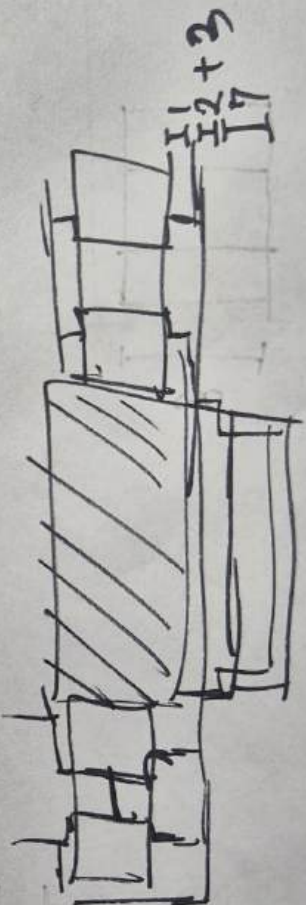
$$3+1+8+1 = 13m$$

$$17.6 (3 \frac{1}{2} - 1)$$





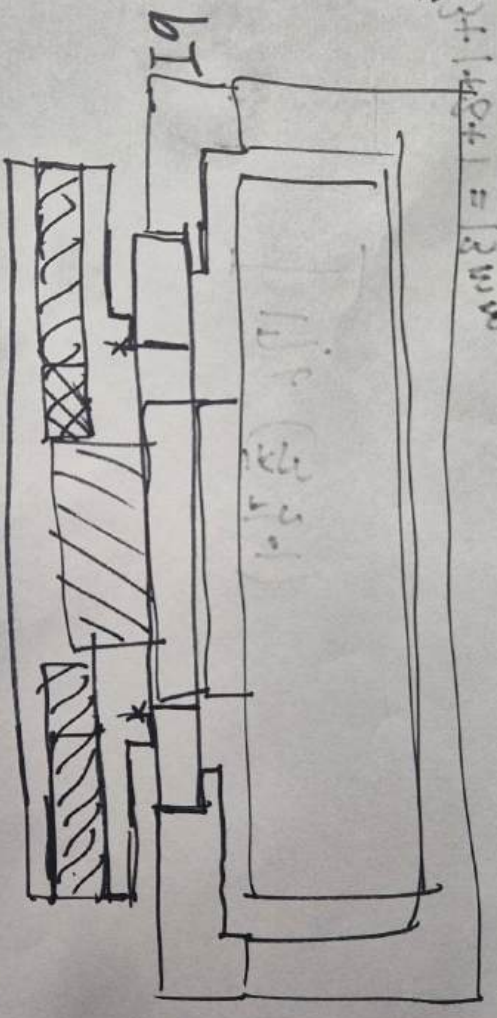
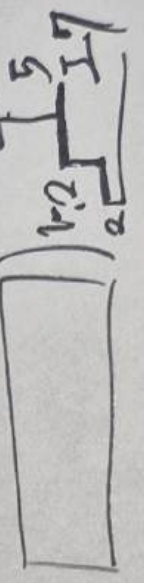
12



4 + 2 + 2



3 + 1.48 + 1 = 5.48



3 + 5 + 1.48 = 9.48

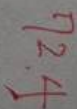
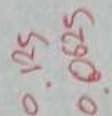
14.54

14.54

- ① 14.54 - 14.54: 12, 14.54: 12.42, 14.54: 14.42, 14.54: 9.92
- ② 14.54 - 14.54: 30, 14.54: 31.06, 14.54: 33.06, 14.54: 28.56
- ③ 14.54 - 14.54: 72, 14.54: 74.54, 14.54: 72.54, 14.54: 77.04



21.74

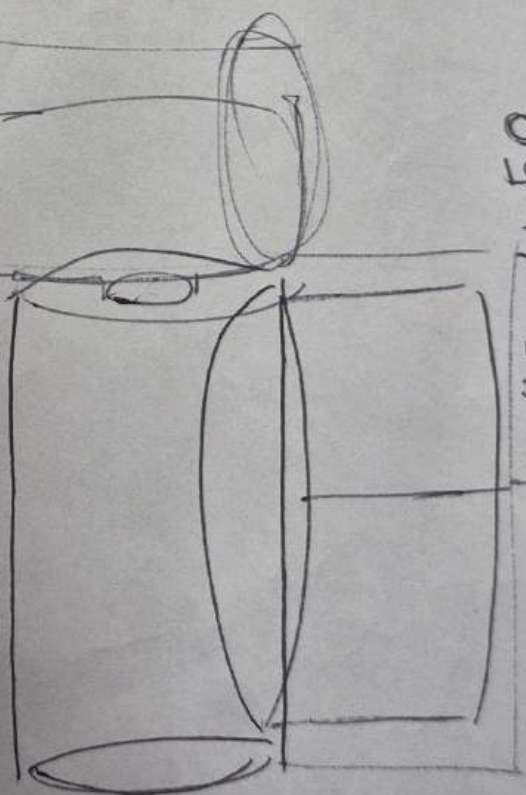


1618

42

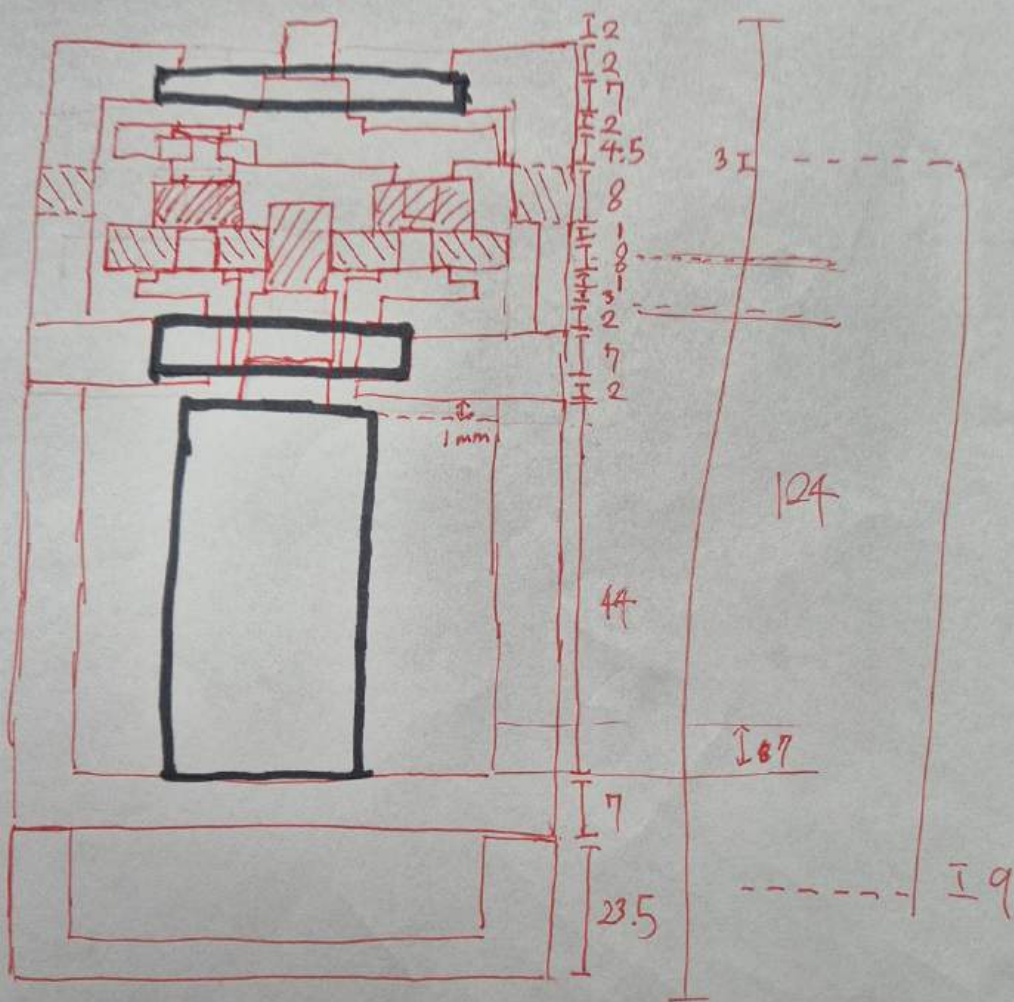
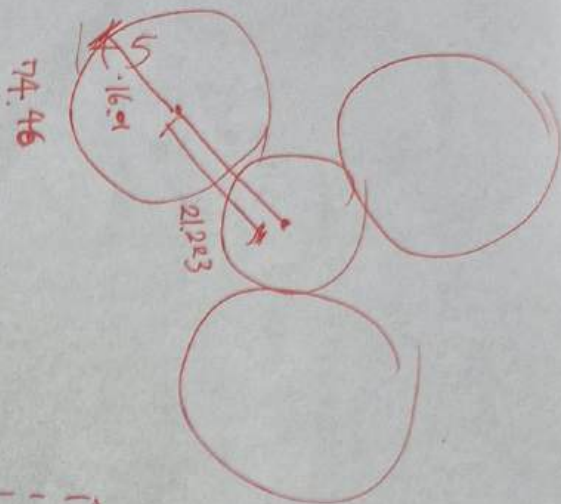
$$\begin{array}{r} 0.125 \\ \hline 23.5 \\ + 50 \\ + 9 \end{array}$$

0.125

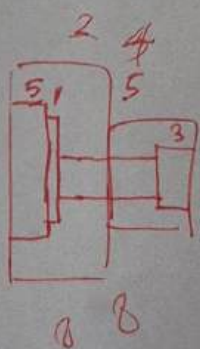


$$123 \overline{) 615} + 50$$

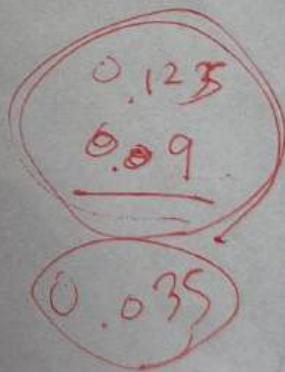
14.06



3.5
1.5 = 5



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0.0625
0.045
0.017

6.87